



Project Outlines

Course and Project Information			
Course Name	System analysis and design	Course Number	0604334
Year	2025/2026	Semester	☒ First ☐ Second ☐ Summer
Credit Hours	3	Project Type	☐ Practical ☐ Theoretical ☒ Combined
Number of project Phases	3	Project Work Format	☐ Individual ☒ Group

Project Objectives		
No.	CLOs	Key Skill
01	Describe the seven stages of the SDLC and differentiate it from the agile and object-oriented methodologies.	Differentiate between the three methodologies
02	Understand the differences between the data flow diagram levels.	Differentiate between the levels and the errors of data flow diagram
03	Understand the designing outputs technologies and human-computer interaction styles	Use the tools for and technologies for designing interfaces.
04	Develop the systems graphically by using Data Flow Diagrams(DFD) and Use Cases Models (UML) that are useful to Design website and smartphone screen .	Draw the DFD level 1 and level 0 Draw and differentiate physical and logical DFD. Use case model.
05	Draw Gantt charts and network diagrams for many projects.	Draw GANTT and network diagram for the project.
06	Implement function point analysis to estimate software projects and PERT calculation to break down projects for better time management.	PERT calculation. Function points. Design (UML and DFD). Create event response table.



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Tools to Use	
Purpose	Tool Options
design	Drawio, visual paradigm, visio
Pert calculation	Ms project
Function point analysis	FP CALCULATOR

Project Theme			
Project Summary	This project qualifies students to apply theoretical knowledge of system analysis and design in a practical, phased, real-case project. Students gradually develop skills by carrying out basic system analysis (problem definition and analysis) throughout the system life cycle. In addition, the students are required to depict the project graphically using context-level data flow diagrams, use cases, and use case scenarios. finally, the student will be asked in this project for performing function points for estimation and pert calculation for scheduling.		
Project Tasks	Phase	Tasks:	Mark
	Phase 1	- Determine the problem. - Determine the scope. - Define the functional and nonfunctional requirements. - Define the stakeholders. - Define the objectives. - Define the methodology used to perform the project. - Perform function points for estimation.	5
	Phase 2	- Draw the network diagram. - Draw the Gantt chart. - Identify the critical path. - Draw logical and physical DFD. - Create Event Response Table for the project.	5
	Phase 3	- Project presentation - Project discussion	10



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Deliverables			
No.	Deliverable	Standard	Format
1	Phase 1 report	MS word	.docx
2	GANTT chart, network diagram, logical and physical diagram, PERT analysis and event response table	Visio, visual paradigm, ms project.	.mpp, .vsdx, .vpp
3	Project presentation	Power point	pptx

Learning Resources	
Textbook(s)	Kendall, K. E., Kendall, J. E., Kendall, E. J., & Kendall, J. A. (2014). Systems analysis and design (Vol. 9). Englewood Cliffs, NJ: Prentice Hall.
References	1. Modern Systems Analysis and Design By: Jeffrey A. Hoffer, Joey F. George and Joseph S. Valacich Printed in the US....2nd edition to 6th edition, 2002...2010 by Pearson Education, Inc. ISBN: 0-8053-2499-2 2. systems Analysis & Design (3rd Edition) By: Alan Dennis, Barbara Haley Wixom, 2. Roberta M. Rogh Printed in the US 2006 by John Wiley & Sons Inc., ISBN: 13:978-0-471-72257-1 or ISBN: 10:0-471-72257-x { See: www.wiley.com/college/dennis}
Recommended Readings	
Online / Electronic Resources	

Submission Details	
File Naming Format	
Submission Platform	☒ Moodle ☐ MS Teams
Submission Deadline	Date: _____ Time: _____